

Luis Guzman-Cornejo

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EDUCATION

University of Oregon, Eugene OR

September 2022 – June 2026

Bachelor of Science

Major in Computer Science; Minor in Mathematics

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Bash, Lua, Java

Graphics/Systems: OpenGL, CUDA, UNIX, Git

Frameworks/Tools: Flask, Unity, MongoDB, Docker

EXPERIENCE

Computer Science Teaching Assistant

September 2023 – June 2024

- Supported and mentored students in Python and C assignments through weekly office hour sessions
- Reviewed and contributed to grading scripts and key course assessments for release to students

LEADERSHIP & PROJECTS

InizioEngine – C++, OpenGL

November 2025 – Current

- Developed a custom game engine from scratch in C++ and OpenGL, implementing a real-time rendering pipeline: shader compilation, vertex buffer management, texture mapping, and transformation pipelines
- Built core runtime systems including player movement, physics and collisions, and animation state machine driven by matrix transformation logic

Software Rasterizer – C++

September 2025 – November 2025

- Built a software-based triangle rasterizer handling pixel rendering without GPU acceleration
- Implemented the full 3D transformation pipeline (model, view, projection) alongside scanline drawing and depth buffering from scratch

Project Rhamnusia – C#, Unity

February 2026 - Current

- Independently building a 3D action-RPG in Unity
- Built an asynchronous scene-streaming system that loads and unloads world chunks around the player using neighbor prefetching and hysteresis-based unloading, eliminating traversal frame hitches and bounding runtime memory
- Engineered data-driven combat and finite-state enemy AI with NavMesh pathfinding and vision-based detection
- Architected the player system as a hierarchical state machine over a decoupled movement layer, keeping systems modular and easy to extend

Carcosa – C#, Unity

May 2026

- Led a four-person team to a 1st-place finish at QuackHacks 3, shipping a Unity game in under 24 hours as gameplay and technical lead
- Designed the core gameplay and architected the project's systems; mentored a teammate with no Unity experience through the engine's toolset and collaborated closely with the team's lead artist on visual direction

AWARDS

BSides PDX – 1st place, CTF competition, specialized in binary reverse engineering with Ghidra October 2025

QuackHacks 3 – 1st Place, Pipeworks Game Design Track May 2026